Ascot Micro-Forest Connect

Loyola Marymount University

WILLETTE LAB OF APPLIED ECOLOGY

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What is the Ascot Micro-Forest Project?

1. Planting Micro-Forests

2. Building Software Infrastructure

3. Expanding Education and Impact

4. Engaging Communities

Project Components

Data Contribution Mobile App

For contributors: A mobile app that allows researchers and visitors alike to record plant growth for individual micro-forests.

Steward Portal

For stewards, or micro-forest administrators: a portal that allows them to view statistics at a glance and manage contributors and data.

Micro-Forest Connect Interface

For curious observers: an interactive page that shows micro-forest locations and info on a dynamic map, with quick facts demonstrating impact!

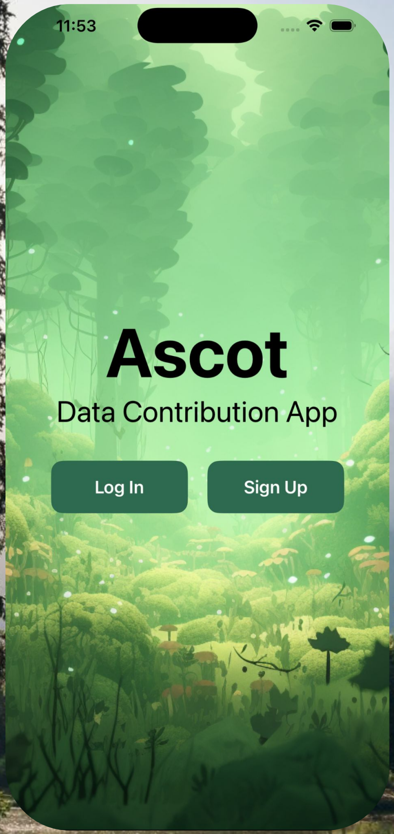
Project Goals

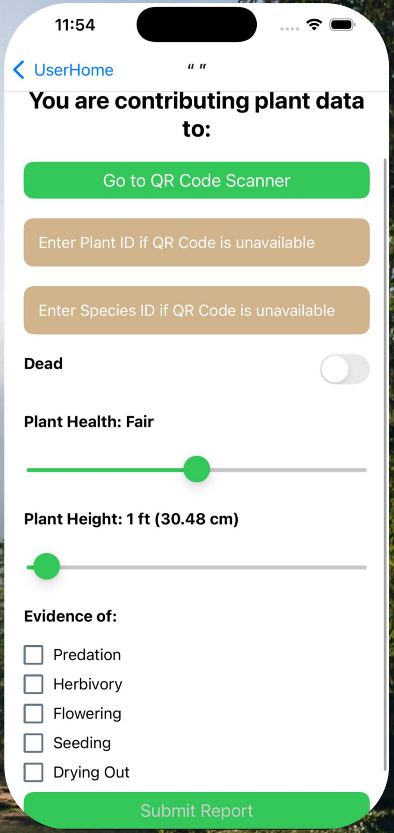
- → Make contribution easier than ever
- → Allow new micro-forests to join on
- → Give stewards control of their data
- → Improve reach and visibility
- → Modernize and prepare the project for scale

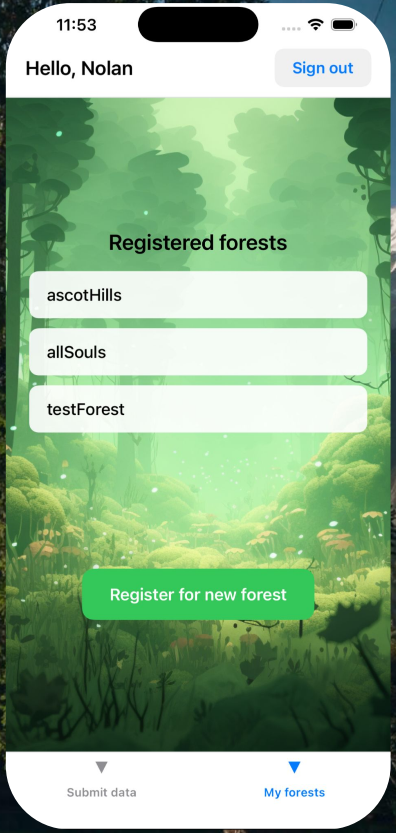
Motivation - Why Ascot?

Concept and Subject Matter	- Allowing communities to see how much they're benefiting the environment - Encourage the proliferation of micro-forests around the LA area and the world
Interdisciplinary Connection	- A chance to collaborate outside of the computer science department - Combines our technical expertise with the subject matter of another department
Project and Codebase Format	- Codebase is in a familiar format, we can leverage past experience with full-stack engineering - A good chance to exercise communicating with a "customer"/client

User Interface



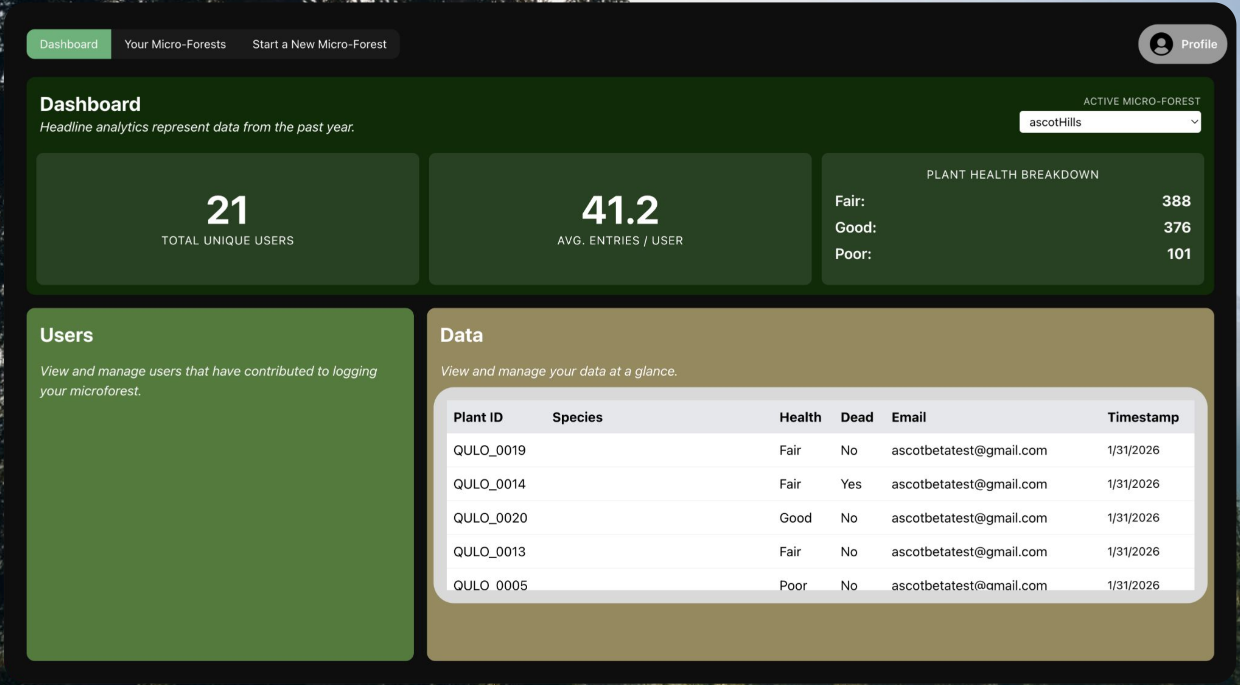


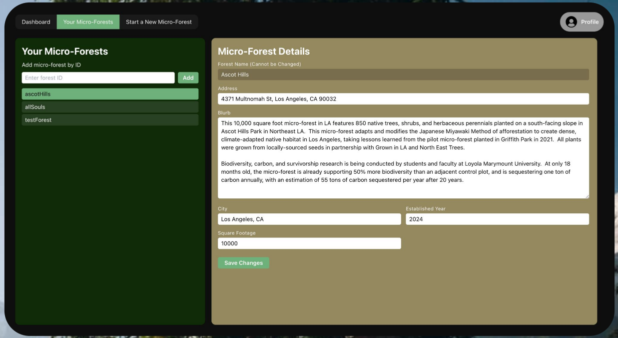


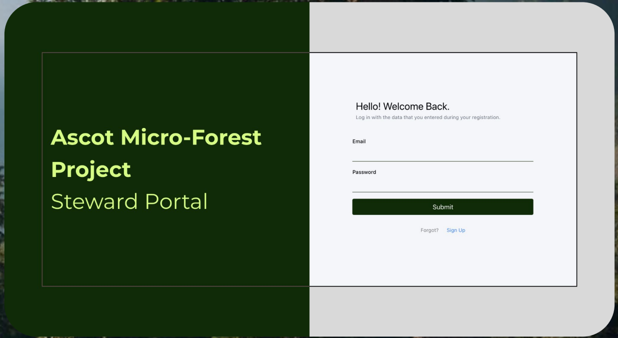
Data Contribution

User Interface

BEFORE: Steward Portal

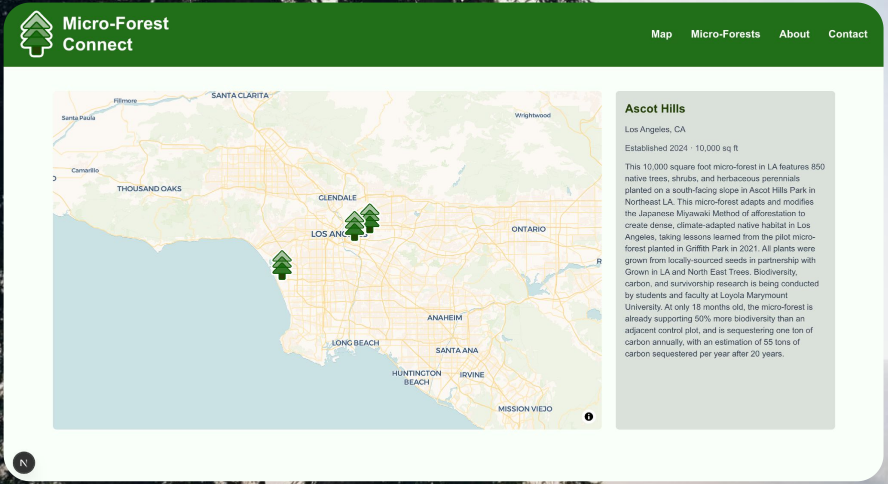


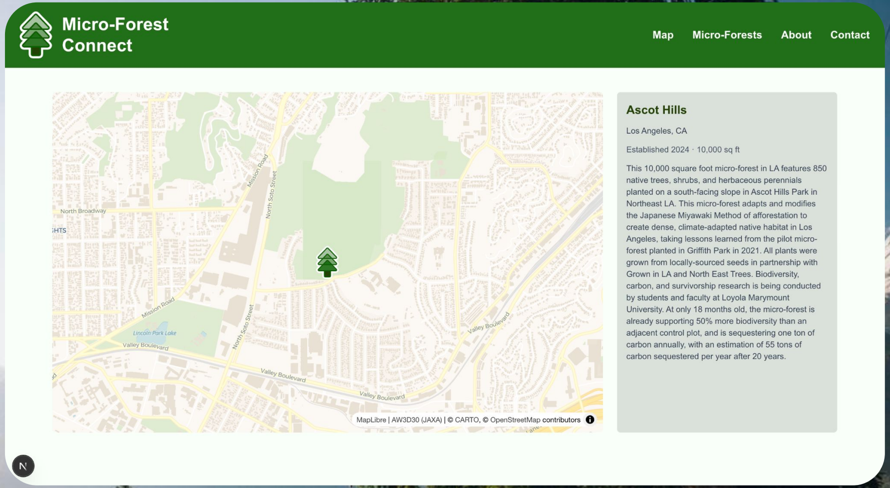


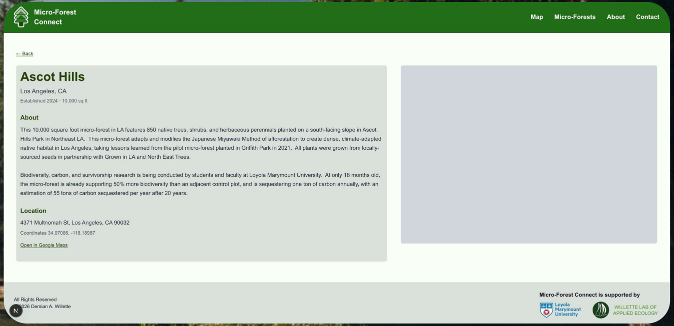


User Interface

Steward Portal







User Interface

Micro-Forest Connect

Demonstration

Project Challenges

Managing 3 Different Types of Users	- Ascot involves 3 user bases with different priorities: Contributor, Steward, Observer - Decisions on security, accessibility, features, and more are decided separately per-component
Metric Generation from Raw Data	- We want to calculate complex statistics like CO_2 absorption from our data - But, have to start with rudimentary data points like plant height, diameter, and health indicators
Codebase Modernization	- The original codebase was in need of updating, to new versions and new technologies altogether - All three components use different developer libraries to be deployed and hosted, while all hooking up to the same database

Status Update & Punch List

Data Contribution App

- The Data Contribution App is already online and ready to serve!
- New update will allow plants to be tracked in separate databases representing different micro-forests, to prepare for scale in the Ascot Project
- Forests can be joined using a verification code

Steward Portal

- Steward Portal has been completely overhauled since beginning the project, from a web app stub to an informative dashboard for micro-forest managers
- Currently working on user flow for creating a new micro-forest, allowing Dr. Willette to approve requests to join the project

Micro-Forest Connect

- The Micro-Forest Connect Interface is the newest component in the project
- Currently has an interactive map showing forest locations and quick facts
- Working on compiling plant data into statistics that track impact, such as CO2 absorption rate

Punch List:

- Finish Create New Micro-Forest flow so that Dr. Willette can approve instantiation of new micro-forests
- Make blog/editorial components of Micro-Forest Connect editable by Dr. Willette without having to code
- Ascot will continue to evolve and grow – write up documentation and jumping-off points for future devs

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Any questions?

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